Recovering Motion-Referring Expressions in Moving-Camera Videos via Global Motion Compensation

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Abstract

Referring multi-object tracking takes a video and a natural-language expression and tracks every object the expression describes. Many expressions refer to how an object moves rather than how it looks, but common hosts read motion from raw box displacement. A moving camera corrupts that signal: a parked car sweeps across the frame and is scored as "moving," while a car keeping pace with traffic can read as static and be dropped. Motion-language RMOT fuses dynamics with text yet leaves the camera in the signal; ego-motion compensation removes the camera but is used only for association. We connect the two with a decision-level plug-in: per-frame homographies compose into a cumulative transform, residual velocity is read at multiple gaps, a two-tower encoder aligns it with the expression, and the score is added to the host logit without host retraining. On iKUN, aligning motion with language lifts moving-class HOTA by +8.83 (native 20.25 → 29.08, n=5); most of that lift comes from giving the host any motion descriptor, while ego-compensation is the only engineered component with a significant effect and the only one that serves both directions of the disambiguation: removing it costs −1.77 moving (p=0.006) and −2.07 static-class HOTA ($p<10^{-4}$), the parked objects that raw velocity misreads as moving. Across three host–split settings, the recovered signal is monotone-inverse to native motion ability. In pooled HOTA the module matches iKUN's published score and improves FlexHook's V2 score outright (+0.281), while FlexHook V1 remains below its published number under a known reproduction gap.

Keywords

referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

1 Introduction

Referring multi-object tracking (RMOT) outputs trajectories for the objects described by a free-form language. For example, a language description such as *left-vehicles-in-red* in the Refer-KITTI dataset is used to evaluate RMOT performance. Many of the descriptions in the Refer-KITTI dataset focus on object motion, such as "moving cars" or "a vehicle turning left", rather than appearance. However, in videos captured by a moving camera, the observed motion is a mixture of object motion and camera motion. As a result, a stationary car may appear to move because of camera motion, while a moving car may appear stationary if it moves together with the camera.

VMRMOT [13] and its RMOT lineage [4, 10, 21] align motion with language. However, they do not explicitly distinguish object motion from camera motion. Conventional multi-object tracking (MOT) addresses this problem through camera motion compensation. For example, BoT-SORT [1] estimates camera motion, while UCMCTrack [22] stabilizes object trajectories. However, the compensated motion is not incorporated into language-guided tracking. Camera motion can be obtained from the global positioning system (GPS) and inertial measurement unit (IMU) annotations provided by KITTI or estimated from image-based methods such as ground-plane projection. Existing RMOT methods, however, do not use this information for language-guided tracking. Instead, our method estimates camera motion directly from image features, eliminating the need for GPS or IMU sensors. This design also allows the module to be integrated into different two-stage RMOT frameworks without modifying their architectures. The only additional requirement is a single scalar calibration for each host model, as described in Section 3.4.

Global Motion Compensation-Link (GMC-Link) separates camera motion from object motion to provide motion information for language-guided tracking. It consists of four stages. First, a homography, which describes the geometric transformation between consecutive frames, is estimated to measure camera motion. The estimated homographies are accumulated over multiple frames to compensate for camera motion and provide a stable reference for measuring object motion. Second, residual velocity, which represents object motion after camera motion has been removed, is computed over multiple frames and used as a temporal motion descriptor. Third, the Contrastive Motion Aligner compares the motion descriptor with the language expression and produces a motion matching score. Finally, the motion score is combined with the host model through a calibrated decision-level correction, allowing GMC-Link to improve existing two-stage RMOT frameworks without modifying their architectures.

The accumulated homographies are the key component of GMC-Link. By compensating for camera motion, they allow the motion descriptor to represent the object's movement rather than the camera's movement. They also preserve motion information across multiple frames, enabling the model to capture motion patterns described by expressions such as "moving" or "turning". Based on this design, we hypothesize that camera motion compensation is essential for distinguishing object motion from camera-induced motion. If residual velocity is replaced with raw image velocity, performance should decrease for moving objects, while stationary objects are more likely to be misclassified as moving. Section 4 evaluates this hypothesis and shows that residual velocity contributes most of the improvement for moving objects, while camera motion compensation makes the motion descriptor effective for object disambiguation rather than simply providing additional motion cues.

This work connects ego-motion compensation with motion-language modeling for referring multi-object tracking (RMOT). Our main contributions are as follows.

- We propose Global Motion Compensation-Link (GMC-Link), a plug-in module that compensates for camera

motion without modifying or retraining the host RMOT model.
- We introduce a motion representation based on residual velocity after camera motion compensation for motion-language alignment.
- We provide experimental evidence showing that camera motion compensation is the key factor for improving motion-aware RMOT.
- We demonstrate consistent improvements on iKUN and FlexHook while preserving the original detector, tracker, and appearance-language matching.

2 Related Work

2.1 Referring MOT and motion-language fusion

TransRMOT [21] established RMOT on Refer-KITTI, iKUN [4] decoupled grounding from association through a frozen tracker, and FlexHook [10] strengthened the two-stage cross-modal hook. The two-stage referring-by-tracking line continues to grow, with zero-shot MLLM captioning over frozen tracks (ReferGPT [3]) and memory-efficient plug-in referring modules (MEX [19]). Later RMOT methods improve semantic grounding [7, 8, 11, 14]; VM-RMOT [13] even makes motion explicit with position, direction, distance-trend, and speed-trend descriptors. The common limitation is where the dynamics come from: raw image-plane boxes. Thus motion-language RMOT can be language-aware while its motion representation remains entangled with camera ego-motion. TempRMOT [23] already aggregates trajectory history in recurrent memory, and we exclude such hosts on evidence rather than convenience: cascading the module onto TempRMOT regressed pooled HOTA in two independent trials (−6.75 under an early cascade rule, −3.8 to −5.4 across a later coefficient sweep), consistent with an external motion module adding redundant constraints to a host whose recurrent state already encodes trajectory history.

2.2 Camera and ego-motion compensation in tracking

Camera compensation is standard in plain MOT, where it stabilizes predictions before association. BoT-SORT [1] estimates an inter-frame transform from ORB matches [17] with RANSAC [5], while UCMCTrack [22] uses a ground-plane camera-motion model. EMAP [15] goes furthest in this direction, decoupling camera rotation and translation from object motion inside the Kalman prediction itself and improving KITTI HOTA by over 5% across four trackers. All three improve association under ego-motion, but the corrected geometry is consumed immediately by frame-to-frame association. It is not composed into a persistent ego-stable trajectory, accumulated over the temporal span of a motion expression, or exposed as a language-aligned feature. The geometric core we rely on is classical — plane-plus-parallax motion separation [9], homography estimation [6], motion segmentation under a moving camera [2], and ego-motion estimation in visual odometry [18] — and we claim no novelty in the separation itself; the contribution is routing its output to language.

2.3 The unfilled intersection

The gap is specific: motion-language RMOT is language-aware but camera-contaminated; camera-compensated MOT is ego-stable but language-unaware. GMC-Link connects them by producing persistent ego-stable motion, aligning it with the expression, and fusing it after the host score. The method therefore preserves host appearance grounding while supplying the motion evidence that the host never had.

3 Method

Figure 1 shows the overall architecture of GMC-Link. The module is attached to an existing two-stage RMOT framework after object detection, tracking, and appearance-language matching have been completed. Instead of modifying the host model, GMC-Link estimates camera motion from consecutive frames, derives residual object motion, aligns it with the language expression, and produces a motion score. This score is then combined with the host model's original prediction through a calibrated decision-level fusion. The complete pipeline consists of the four stages shown in Figure 2.

3.1 Ego-motion compensation

The first stage estimates camera motion between consecutive frames. We extract and match background keypoints using Oriented FAST and Rotated BRIEF (ORB) [17]. Keypoints inside tracked object boxes are removed before matching so that moving foreground objects do not bias the camera-motion estimate. The remaining correspondences are used to estimate a homography $H_{t-1\to t}$ with Random Sample Consensus (RANSAC) [5], representing the geometric transformation from frame I_{t-1} to frame I_t caused by camera motion.

To describe object motion over multiple frames, the estimated homographies are accumulated as

$$H_{t-k\to t} = H_{t-1\to t}\, H_{t-2\to t-1} \cdots H_{t-k\to t-k+1}, \tag{1}$$

where $H_{t-k\to t}$ represents the accumulated camera transformation from frame I_{t-k} to frame I_t. The accumulated homography is applied once to the original object centroid, rather than repeatedly transforming already warped coordinates, to reduce numerical drift and provide a stable reference for residual-motion extraction.

3.2 Multi-scale residual velocity

After ego-motion compensation, object motion is represented by residual velocity, which removes the estimated camera motion from the observed object motion. For a temporal gap g, the raw velocity is the displacement of the object centroid between two frames and contains both object motion and camera-induced motion. The ego velocity represents the displacement caused by camera motion alone, which is obtained by applying the accumulated homography in Eq. 1 to the previous object centroid. The residual velocity is computed as

$$v_g^{\mathrm{res}} = v_g^{\mathrm{raw}} - v_g^{\mathrm{ego}}, \tag{2}$$

where v_g^{res}, v_g^{raw}, and v_g^{ego} denote the residual, raw, and ego velocities, respectively. This representation approximates the motion that would be observed by a stationary camera. For example, a parked vehicle has near-zero residual velocity even when camera

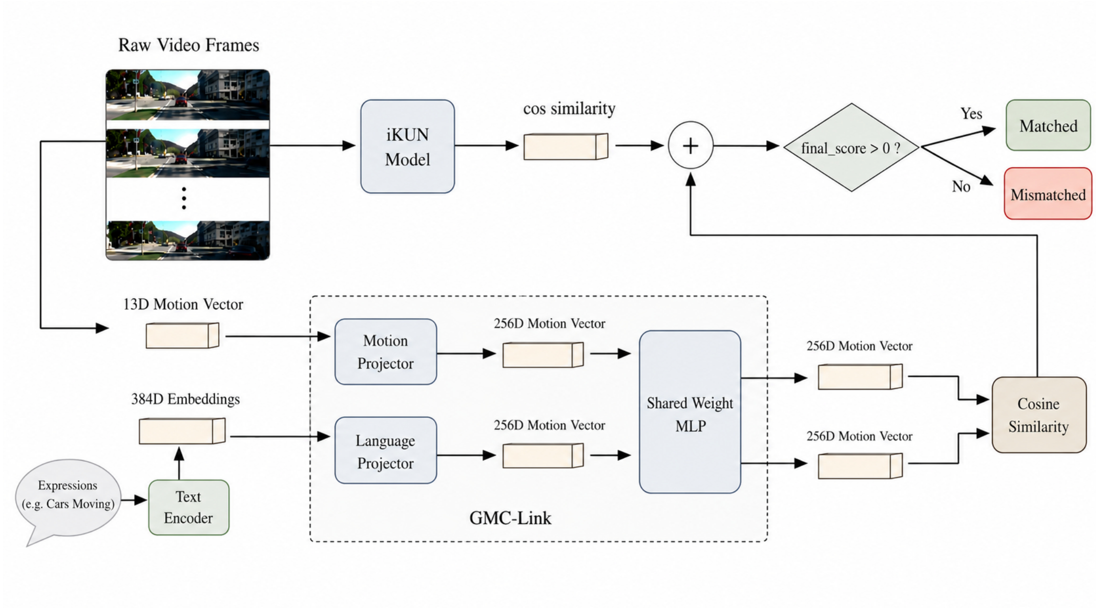

Figure 1: GMC-Link architecture. The host provides tracks and expression scores; GMC-Link estimates camera motion, derives residual motion, aligns it with language, and returns only a calibrated decision-layer correction.

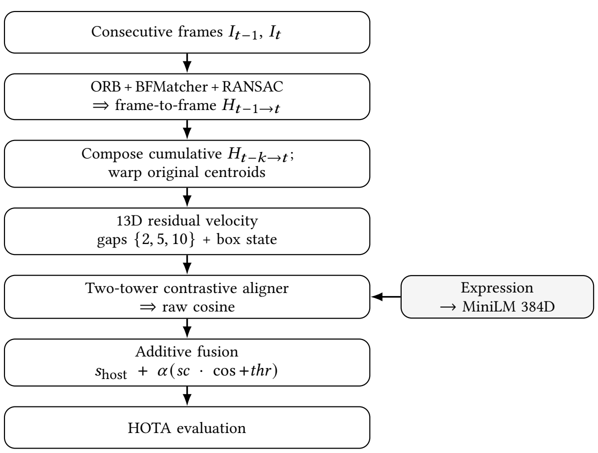

Figure 2: GMC-Link pipeline. ORB homographies compose into a cumulative transform, produce a 13D residual-velocity descriptor, align with the expression, and correct the host logit for HOTA evaluation.

motion causes large image displacement. All velocities are normalized by the image dimensions to make them independent of image resolution.

To capture motion over different time spans, residual velocities are computed at temporal gaps of $\{2, 5, 10\}$ frames. The residual velocities are concatenated with the object box state to form a 13-dimensional motion descriptor,

$$\big((dx_s, dy_s), (dx_m, dy_m), (dx_l, dy_l), \Delta w, \Delta h, c_x, c_y, w, h, snr\big), \quad (3)$$

where the first six values are the residual velocities at the three temporal gaps. The remaining features describe the object box size and location. The final feature, signal-to-noise ratio (SNR), measures the reliability of the estimated camera motion. A larger SNR indicates more reliable ego-motion compensation and provides a more stable motion representation for the following motion-language alignment stage.

3.3 Motion-language alignment

The motion-language alignment stage measures the similarity between the residual motion and the language expression. The 13-dimensional motion descriptor is encoded by a motion encoder, while the language expression is encoded by all-MiniLM-L6-v2 (MiniLM) [16] into a text embedding. Both embeddings are projected into the same feature space using linear projection layers and a shared Multi-Layer Perceptron (MLP).

The model is trained using Information Noise Contrastive Estimation (InfoNCE) [20] loss to maximize the similarity between matched motion-expression pairs while separating unmatched pairs. During inference, the cosine similarity between the motion and language embeddings is used as the motion matching score. This score is passed to the decision-level fusion stage described in Section 3.4.

3.4 Decision-level additive fusion

The motion matching score is fused with the host prediction at the decision level:

$$s_{\text{final}} = s_{\text{host}} + \alpha \left(sc \cdot \cos + thr\right), \quad (4)$$

where s_{host} is the host score, cos is the motion-language cosine similarity, sc rescales the motion score, thr shifts the decision threshold, and α controls the motion contribution.

Table 1: Locked fusion coefficients for Eq. 4, hand-set per host–split setting. FlexHook (V1) and (V2) are one architecture calibrated per split. Appearance scale is damped because motion is uninformative on appearance expressions.

Host	Axis	α	*sc*	*thr*
iKUN	motion	1.00	0.90	+0.17
	appearance	1.00	0.30	+0.10
FlexHook (V1)	motion	0.65	10.0	+3.0
	appearance	1.00	3.50	+0.9
FlexHook (V2)	motion	0.40	10.0	+1.3
	appearance	1.00	3.50	+1.2

Calibration parameters are selected separately for each host model because RMOT scores have different ranges; the values used in this work are listed in Table 1. During inference, a keyword-based router sends motion-related expressions, such as *moving*, *turning*, *parked*, and *braking*, to the motion branch and all other expressions to the appearance branch.

Because GMC-Link only modifies the final score, it can be integrated into existing two-stage RMOT frameworks without changing or retraining their detector, tracker, visual encoder, or language encoder.

4 Experiments

4.1 Setup

We evaluate on Refer-KITTI V1 [21] and V2 [23] using TrackEval HOTA [12]. Moving-class HOTA tests whether the module fixes the target failure; pooled HOTA checks whether that recovery survives the full benchmark. Comparing iKUN [4] with FlexHook [10] tests whether the gain concentrates where native motion reasoning is weakest, and the raw-velocity ablation tests whether ego-compensation matters beyond adding any motion descriptor. FlexHook (V1/V2) is one architecture calibrated per split, not two independent methods. Results are mean±std over n=3 seeds for cross-setting tables and n=5 for ablations with Welch tests; a seed randomizes aligner training (initialization and batch order) only, while homography estimation and host outputs are deterministic. The aligner uses AdamW (10^{-3}, batch 256, 100 epochs); ORB uses 1500 features and RANSAC a 5 px threshold. All iKUN comparisons use the paper-canonical ground-truth template aligned with the public NeuralSORT detections, and each host runs its published public detections unchanged. We report native hosts and a simple uncalibrated raw-cosine GMC control to separate adding the signal from calibrating it to the host logit scale. Code, trained aligner weights for every seed, per-sequence score caches, and the coefficient tables will be released upon publication.

Refer-KITTI has no held-out tuning split, so Table 1's shipped coefficients are calibrated on the benchmark sequences and should not be claimed as universally transferable hyperparameters. Two observations bound the fitting capacity. First, the scale coefficients were calibrated independently per FlexHook split yet came out identical (*sc*=10.0 motion, 3.5 appearance on both V1 and V2 in Table 1), suggesting the calibration tracks the host's logit range rather than the particular test sequences; with those shared, roughly four fitted scalars per setting remain against three to four evaluation sequences. Second, we use leave-one-sequence-out calibration to bound overfitting risk directly. iKUN LOSO moving HOTA remains 28.38 ± 1.60 versus 29.47 ± 0.88 in-sample on the same three seeds (native 20.25; the n=5 ablation reference in Table 4 is 29.08 ± 0.83), while FlexHook LOSO pooled is 53.28 ± 0.17 (V1) and 42.74 ± 0.07 (V2), with V2 still above the published 42.526. We therefore claim the iKUN moving-class recovery and the FlexHook V2 pooled gain, not the iKUN pooled number, as calibration-robust.

The test splits are heavily skewed toward appearance. V1 holds 158 expressions over 1010 frames, of which 75.3% are appearance queries, 7.6% are static-motion queries, and only 17.1% (27 expressions) refer to a moving object; V2 holds 862 expressions over 1469 frames with a similar 74.1 / 11.7 / 14.2 split. The motion class our module targets is therefore a roughly one-in-six minority. Pooled HOTA is computed once over the union of all trajectories, so it is dominated by detection and association counts from the appearance majority; it is not a class-weighted mean of per-class HOTA. A recovery concentrated on moving expressions can therefore register as a small pooled movement even when the per-class effect is large. The module's target failure is motion-class grounding, so moving-class HOTA is the pre-specified effect metric rather than a post-hoc selection; pooled HOTA is the field-standard whole-benchmark summary, reported for every setting with all deltas anchored to the native host.

4.2 Main results

Table 2 first shows that raw GMC evidence is not sufficient by itself: the uncalibrated *+GMC simple* control is negative on iKUN and FlexHook V1. The signal must be calibrated to the host's score range. With the shipped calibration, iKUN [4] reaches 44.634±0.066, matching rather than materially beating the published 44.564, while FlexHook V2 gives the clean pooled improvement, 42.807 ± 0.038 versus 42.526 (+0.281). Pooled parity on iKUN should be read as a guarantee rather than a shortfall: the moving-class recovery is purchased at no cost to the roughly 83% of queries the module does not target, which is the property that makes a plug-in safe to attach. FlexHook V1 improves over *+GMC simple* (53.121 $\rightarrow$ 53.526) but remains below its published 53.824: no tested configuration beats that number — the strongest legacy variant reaches 53.716 ± 0.068 (n=3) — so we treat the V1 gap as a reproduction limit of our pipeline and report the module's V1 gain against the *+GMC simple* control. The scientific claim is therefore not that GMC-Link beats every pooled leaderboard number; it is that ego-compensated motion recovers motion-class errors, especially where native motion reasoning is weakest. For external context, VMRMOT [13], the strongest motion-language host on this benchmark, reports 53.00/35.21 pooled HOTA on V1/V2 with its own end-to-end detector and tracker. The FlexHook settings with our module reach 53.526/42.807 on the same splits, but we attribute that ordering to host and pipeline rather than to the module and do not claim a head-to-head win; the class-level comparison our claim would require is not computable from published numbers, as VMRMOT does not report per-class moving HOTA. We therefore compare each host against its own native score.

Table 2: Pooled HOTA on Refer-KITTI test (n=3, mean±std). Δ is against published pooled HOTA; iKUN matches, while FlexHook V2 is the clean pooled improvement.

Host	Native	+GMC simple	+GMC (ship)	Δ paper
iKUN [4] (V1)	44.564	44.272	44.634 ± 0.066	+0.070
FlexHook (V1) [10]	53.824	53.121	53.526 ± 0.087	−0.298†
FlexHook (V2) [10]	42.526	42.532	42.807 ± 0.038	+0.281

†FlexHook (V1) paper gap is structural (no tested configuration beats 53.824); Δ vs *+GMC simple* anchor = +0.405.

Table 3: Motion deficit across host–split settings. Native moving-class HOTA versus the moving-class HOTA gain added by the module. The gain is monotone-inverse to native motion ability across the three settings; the FlexHook rows are one architecture calibrated per split.

Host	Native MOVING	GMC MOVING gain
iKUN [4]	20.25	+8.83 ± 0.83 (n=5)
FlexHook (V1) [10]	43.98	+0.91 ± 0.79 (n=3)
FlexHook (V2) [10]	48.02	+0.43 ± 0.17 (n=3)

Moving-class denominators: 27 of 158 expressions (V1); 14.2% of 862 (V2).

4.3 The motion deficit

The gain tracks native motion deficit. iKUN starts lowest and gains most (+8.83 ± 0.83); FlexHook V1 and V2 start higher and gain little (+0.91 ± 0.79 and +0.43 ± 0.17). Across the three settings, the relation is monotone-inverse: larger native deficit receives larger correction. This supports the intended use case, a host that already grounds appearance but does not reliably distinguish ego-induced displacement from object motion. We treat the pattern as a prediction over three settings — effectively two independent architectures, since the FlexHook pair shares weights-in-kind — not a proportional law; confirming it as a characterization requires further hosts.

4.4 Ablations

The ablation directly tests the Introduction hypothesis, and we report the decomposition rather than the headline alone. The full module lifts the moving class 20.25 → 29.08 (+8.83); the −ego arm, which feeds raw uncompensated velocity to the same aligner, already reaches 27.31, so having a motion descriptor at all supplies +7.06 of the recovery and ego-compensation the remaining +1.77 (Welch t=3.75, p=0.006, 95% CI [0.67, 2.87], Cohen's d≈2.4). The moving class understates what compensation buys, because raw velocity errs in both directions: on the static class, the −ego arm drops 43.26 → 41.19, a −2.07 cost (t=9.81, $p<10^{-4}$) concentrated exactly on the parked objects whose pixels sweep across the frame. Single-gap motion (−0.26 moving, p=0.64) and removing snr (+0.67 moving, p=0.19) stay within seed spread on both classes, so ego-compensation is the only statistically significant engineered component, and the only one the disambiguation claim rests on; its effect survives Bonferroni correction over the three component tests (p_{corr}=0.018). Two caveats keep the null results honest: at n=5 they bound only large effects, so modest multiscale or snr contributions cannot be excluded, and the −snr arm's higher mean is itself within that spread — snr is retained as the variance-reduction feature of Section 3.2, and dropping it would be a defensible simplification rather than a loss.

Feature-level injection regressed an early host by −21.7% F1, supporting the decision to correct logits after appearance scoring rather than reshaping host representations.

Table 4: Per-class HOTA ablation on iKUN (n=5). Ego-compensation is the only significant engineered component, and the only one that moves both classes: removing it costs −1.77 moving (Welch p=0.006) and −2.07 static ($p<10^{-4}$).

Variant	MOVING HOTA	STATIC HOTA
Native host (no module)	20.25	—
Full module	**29.08 ± 0.83**	**43.26 ± 0.26**
−ego (raw velocity)	27.31 ± 0.66	41.19 ± 0.39
−multiscale (single gap)	28.82 ± 0.84	43.42 ± 0.10
−snr	29.74 ± 0.59	43.35 ± 0.35

Native static-class HOTA under the ablation readout is not archived; deltas among module arms share one convention and are directly comparable.

Figure 3: Real recovery under camera ego-motion on "moving cars" (Refer-KITTI seq 0005, frames 85/94/100 of a forward-driving clip; the green car moves, the red car is parked). The dashed line marks the mover's image column. **Red:** the parked car's centroid slides left as the camera passes it (center-x 261 → 104 px), so the camera-naive host scores it "moving" (false positive), yet its ego-compensated residual is near zero and the module restores it to static. **Green:** the moving car holds its column (center-x ≈605, <1 px/frame), so the host misses it (false negative), but its nonzero residual — unlike the static background — recovers it as moving. Both errors flip from the same host logit through the additive correction.

Figure 3 makes the failure mode concrete. The parked car has large raw image displacement because the camera passes it, while the genuinely moving car maintains nearly the same image column. Raw image-plane motion therefore gives the wrong semantic cue in both directions; ego-compensated residual motion restores the distinction. The module runs at 68 FPS (median 14.8 ms/frame excluding host inference), dominated by CPU ORB homography (14.55 ms); the aligner is under 0.2 ms.[1]

5 Conclusion and Limitations

A decision-level motion-language module is a targeted correction: it recovers the motion-class referring expressions that camera-naive hosts miss, without touching the appearance majority. The bulk of the recovery comes from aligning a motion descriptor with language at all; ego-compensation is the component that turns the descriptor into a disambiguator, significantly protecting both the moving and the static class. The result suggests that motion-language grounding in moving-camera video should reason over ego-stable trajectories, not raw image-plane displacement. The module keeps the tracker and detector fixed, retrains no host weights, and adds only host-specific scalar calibration. On iKUN [4] it matches published pooled HOTA while lifting the hidden motion class; on FlexHook [10] V2 it improves pooled HOTA outright.

The limits are bounded. A single ORB/RANSAC homography [1] is approximate under non-planar scenes and strong parallax; our evidence is therefore confined to planar-dominated Refer-KITTI driving footage, with no claim of transfer to handheld, aerial, or strongly non-planar video. A related failure mode is keypoint starvation: a large mover filling the frame leaves little background texture, triggering the identity fallback exactly when compensation is most needed; the snr feature down-weights such frames but does not recover them. Calibration is host-specific and benchmark-level coefficients are not universal hyperparameters. We study two host architectures across three host–split settings, with FlexHook V1/V2 not independent architectural confirmations. Hosts with native temporal memory, such as TempRMOT [23], remain out of scope on tested evidence: cascaded trials regressed pooled HOTA rather than improving it.

The broader implication is that camera compensation can serve not only tracking association, but also language grounding. Richer ego-stable trajectory descriptors are the natural next step for camera-naive RMOT hosts.

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